PAPER_762: NGC 1792 The Stellar Forge — UQFF Starburst Galaxy Evolution

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Abstract

NGC 1792 (“The Stellar Forge”) is a starburst spiral galaxy located ~42 million light-years away in Columba, forming stars over 10× faster than the Milky Way at ~10 $M_{\odot}$/yr. This paper derives the Master Universal Gravity UQFF equation governing its starburst-driven evolution, incorporating stellar mass growth, supernova feedback suppression, cosmic expansion, and Aether electromagnetic effects. The result g_NGC1792 ≈ 1.053×10^{-2} m/s² is dominated by the [UA]/[SCm] electromagnetic Aether term.

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1. Introduction

NGC 1792 is a classic starburst spiral galaxy with a bright orange core of older stars surrounded by blue young star-forming regions. Its high SFR (~10 $M_{\odot}$/yr) is driven by a large gas reservoir, potentially triggered by past tidal interactions. Supernova feedback from rapid star formation injects energy into the ISM, regulating further star formation over ~100 Myr timescales. The UQFF framework captures these dynamics through non-standard Aether and superconductive magnetism terms.

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2. Master UQFF Gravity Equation

$$g_{NGC1792}(r, t) = \frac{G M}{r^2} (1 + H(z) t)(1 + M_{sf}(t))(1 - F_{sn}(t))(1 + f_{TRZ}) + q(v \times B)\left(1 + \frac{\rho_{vac,[UA]}}{\rho_{vac,[SCm]}}\right) \cdot 10^{-12}$$

2.1 Parameters

Parameter	Symbol	Value	Source
Galaxy mass	M	$10^{10}\ M_{\odot} = 1.989\times10^{40}$ kg	Hubble
Galaxy radius (½ diam.)	r	3.78×10^{20} m (~40 kly)	Hubble
Redshift	z	0.0095	Distance calc
Starburst duration	t	100×10^{6} yr = 3.156×10^{15} s	Labs
Star formation rate	SFR	10 $M_{\odot}$/yr	Hubble

Parameter	Symbol	Value	Source
Feedback amplitude	F_0	0.05	Labs estimate
Feedback timescale	τ_sn	100×10^6 yr = 3.156×10^{15} s	Labs
Orbital velocity	v	10^6 m/s	ISM
Galactic B field	B	10^{-5} T	Labs
ρ_vac,[UA]	—	7.09×10^{-36} J/m³	UQFF
ρ_vac,[SCm]	—	7.09×10^{-37} J/m³	UQFF
f_TRZ	—	0.1	UQFF

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3. Long-Form Derivation

Step 1: Base Gravitational Term

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g_grav = (6.6743e-11 × 1.989e40) / (3.78e20)²
       = 1.328e30 / 1.429e41 = 9.293e-12 m/s²
```

Step 2: Starburst Mass Growth

```
M_sf(t) = SFR × t / M_0 = 10 × 100e6 / 10^10 = 0.1
1 + M_sf(t) = 1.1
```

Step 3: Supernova Feedback Adjustment

```
t / τ_sn = 3.156e15 / 3.156e15 = 1
F_sn(t) = 0.05 × (1 - exp(-1)) = 0.05 × 0.6321 = 0.031605
1 - F_sn(t) = 0.968395
```

Step 4: Cosmic Expansion

```
H(z) = 70 × sqrt(0.3 × (1.0095)³ + 0.7) = 70 × sqrt(1.00861) = 70.301 km/s/Mpc
H(z) = 70.301e3 / 3.086e22 = 2.278e-18 s⁻¹
H(z) × t = 2.278e-18 × 3.156e15 = 7.189e-3
1 + H(z) × t = 1.007189
```

Step 5: Time-Reversal Correction

```
1 + f_TRZ = 1.1
```

Step 6: Electromagnetic [UA] Term

```
q × (v × B) = 1.602e-19 × 1e6 × 1e-5 = 1.602e-18 N
a = 1.602e-18 / 1.673e-27 = 9.575e8 m/s²
(1 + ρ_vac,[UA]/ρ_vac,[SCm]) = 11
Total = 9.575e8 × 11 × 10⁻¹² = 1.053e-2 m/s²
```

Step 7: Final Solution

```
g_NGC1792 = (9.293e-12) × (1.007189) × (1.1) × (0.968395) × (1.1) + 1.053e-2
          = 1.097e-11 + 1.053e-2
          ≈ 1.053e-2 m/s²
```

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4. Physical Interpretation

The starburst phase dominates NGC 1792's evolution. The supernova feedback term ($1 - F_{sn} = 0.968$) captures feedback self-regulation, while the mass growth term ($1 + M_{sf} = 1.1$) reflects the 10% gas-to-star conversion rate over 100 Myr. The Aether [UA] electromagnetic term (1.053×10^{-2} m/s^2) exceeds classical gravity by ~6 orders of magnitude under UQFF, revealing non-standard vacuum energy as a primary driver.

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5. UQFF Framework Advancement

- First UQFF application to a spiral starburst galaxy at 42 Mly
- Supernova feedback modeled as a damping term on effective buoyancy
- Starburst star formation mass growth incorporated as additive UQFF correction
- Demonstrates UQFF scalability to galaxy-scale starburst environments

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6. Conclusions

The Master UQFF gravity equation for NGC 1792 yields $g_{NGC1792} \approx 1.053\times10^{-2}$ m/s^2, confirming the Aether electromagnetic term dominates starburst galaxy dynamics under UQFF. The result encodes the balance between starburst mass growth and supernova feedback suppression over a 100 Myr evolutionary timescale.

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